```
0.3333333

> mean(c(TRUE, TRUE, TRUE))
1
```

Consider the 'Bank Marketing Data Set' for a Portuguese banking institution that is available from the UCI Machine Learning Repository available at *https://archive.ics.uci.edu/ml/datasets/Bank+Marketing*. The data can be used for public research. There are four data sets available, and we will use the *read.csv()* function to import the data from a 'bank.csv' file.

```
> bank <- read.csv(file="bank.csv", sep=";")

> bank[1:3,]
  age         job marital education default balance housing
loan  contact day
1  30 unemployed married   primary      no    1787      no
no cellular  19
2  33   services married secondary      no    4789     yes
yes cellular  11
3  35 management  single  tertiary      no    1350     yes
no cellular  16
  month duration campaign pdays previous poutcome  y
1   oct       79        1    -1        0  unknown no
2   may      220        1   339        4  failure no
3   apr      185        1   330        1  failure no
```

We can find the mean age from the bank data as follows:

```
> mean(bank$age)

41.1701
```

Median

The *median* function in the R stats package computes the sample median. It accepts a numeric vector 'x', and a logical value 'na.rm' to indicate if it should discard 'NA' values before computing the median. Its syntax is as follows:

```
median(x, na.rm = FALSE, ...)
```

A couple of examples are given below:

```
> median(3:5)
4

> median(c(3:5, 50, 150))
[1] 5
```

We can now find the median age from the bank data set as follows:

```
> median(bank$age)

39
```

Pair

The *pair* function can be used to combine two vectors. It accepts two arguments, and vectors 'x' and 'y'. Both the vectors should have the same length. The syntax is as follows:

```
Pair(x, y)
```

The function returns a two column matrix of class 'Pair'. An example is given below:

```
> Pair(c(1,2,3), c(4,5,6))
     x y
[1,] 1 4
[2,] 2 5
[3,] 3 6
attr(,"class")
[1] "Pair"
```

The function is used for paired tests such as *t.test* and *wilcox.test.*

dist

The *dist* function is used to calculate the distance matrix for the rows in a data matrix, and accepts the following arguments:

Argument	Description
x	A numeric matrix
method	The distance measure to be used
diag	Prints diagonal of distance matrix if TRUE
upper	Prints upper triangle of distance matrix if TRUE
p	Power of the Minkowski distance

The distance measurement methods supported are: 'euclidean', 'maximum', 'manhattan', 'canberra', 'binary' and 'minkowski'. The distance measurement for the age data set using the Euclidean distance is given below:

```
> dist(bank$age, method="euclidean", diag=FALSE, upper=FALSE,
p=2)
      1  2  3  4  5  6  7  8  9 10 11 12 13 14 15 16 17 18 19
```